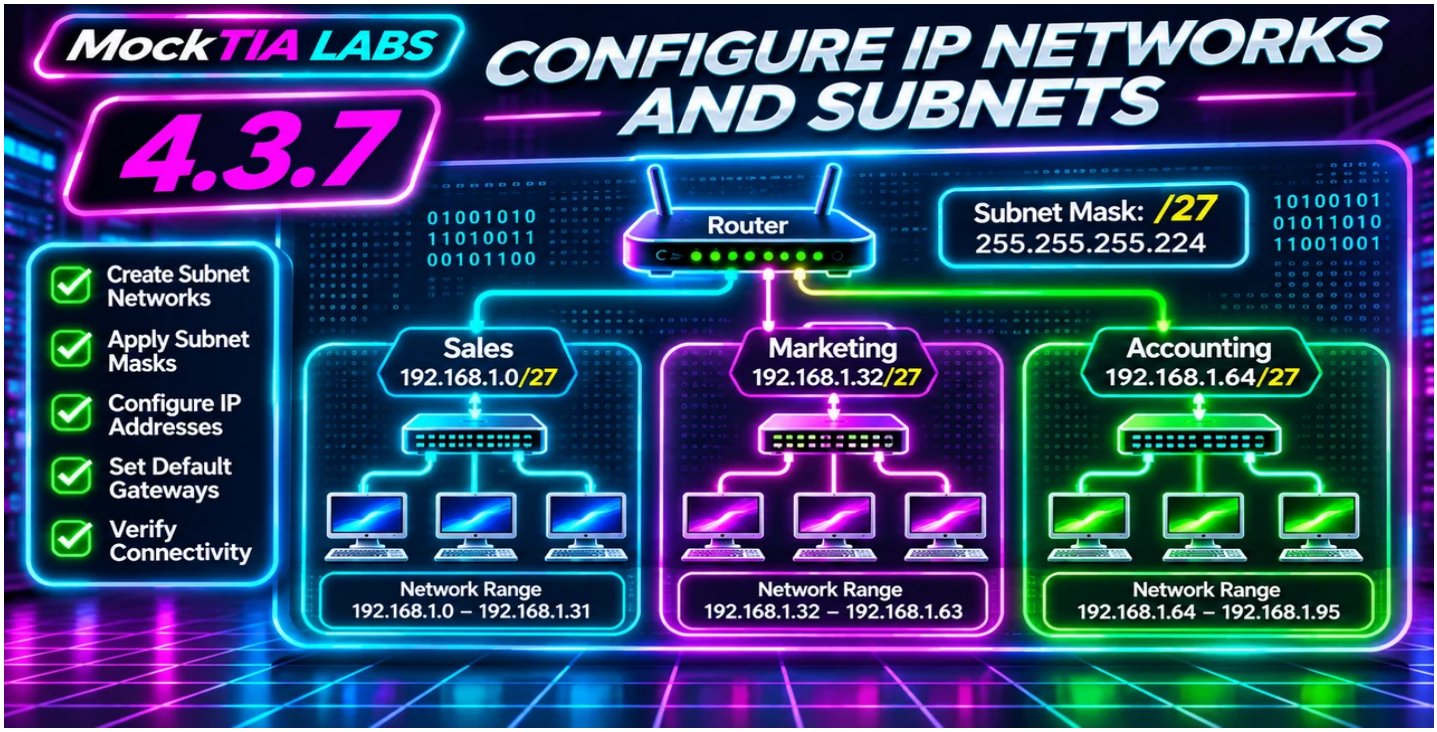
CONFIGURE IP NETWORKS AND SUBNETS

Student Study Guide + Vocabulary + Diagrams



LAB PURPOSE

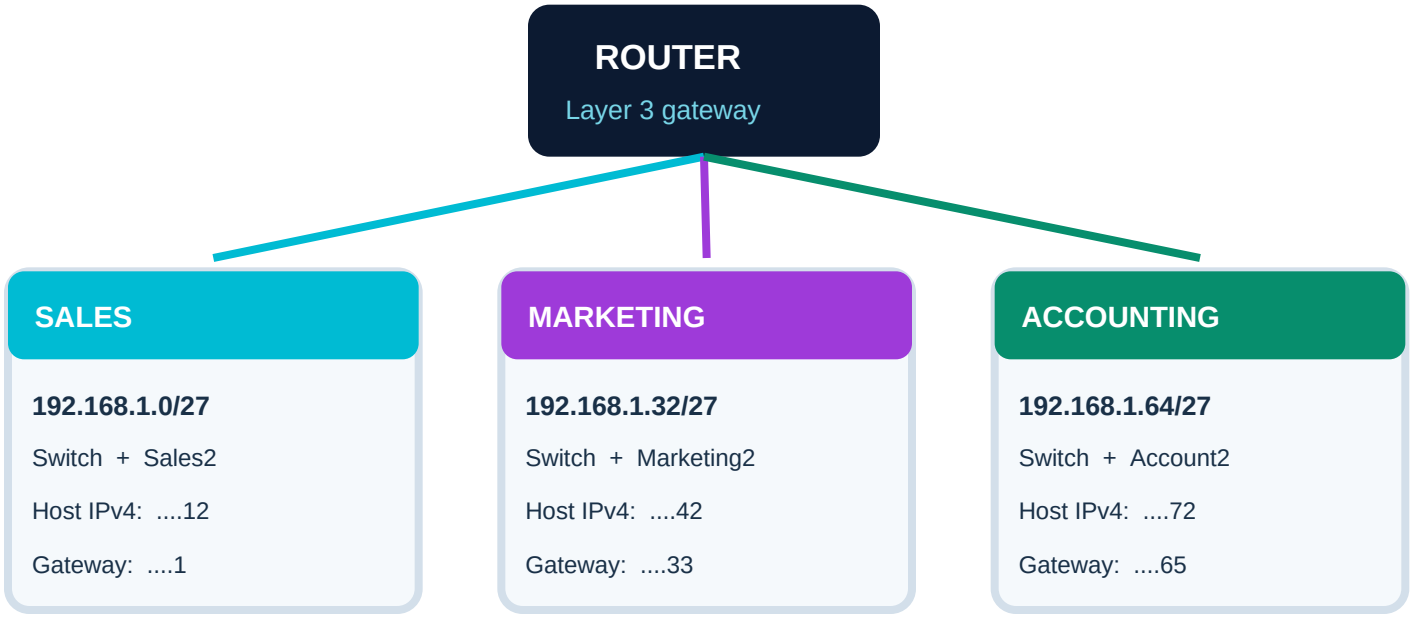
Investigate three Windows workstations using ipconfig, identify /27 network ranges, verify gateways, and explain why hosts on different subnets need Layer 3 routing.

Network: 192.168.1.0/24 | Working subnet mask: 255.255.255.224 (/27)

Device roles: Sales2, Marketing2, Account2, Switch1, Switch2, Router

Three /27 departments

A /27 allocates 32 total addresses per network. The network and broadcast addresses cannot be assigned to ordinary workstations.



IMPORTANT

A switch forwards Ethernet frames but does not perform IP routing. Even with Switch1 linked to Switch2, Sales2 and Account2 are on separate /27 networks. Configure gateways and route between them.

SIMULATOR WORK ORDER

1. Run ipconfig /all on Sales2, Marketing2, and Account2.
2. Determine each host subnet network ID and broadcast.
3. Apply /27 masks and the appropriate default gateways.
4. Link Switch1 and Switch2; compare Layer 2 versus routed traffic.
5. Pass the knowledge check and print your completion PDF.

Understanding /27

Prefix	Subnet mask	Usable hosts
/27	255.255.255.224	30

Subnet ranges for 192.168.1.0/24 split into /27 blocks

Network ID	First usable	Last usable	Broadcast
192.168.1.0	192.168.1.1	192.168.1.30	192.168.1.31
192.168.1.32	192.168.1.33	192.168.1.62	192.168.1.63
192.168.1.64	192.168.1.65	192.168.1.94	192.168.1.95
192.168.1.96	192.168.1.97	192.168.1.126	192.168.1.127

Quick mental math: Block size = 256 - 224 = 32. Subnet starts are 0, 32, 64, 96, 128, 160, 192, 224.

How to calculate a subnet

Step 1: Read the CIDR prefix and convert to a dotted-decimal mask.

Step 2: Calculate the block size from the changing octet.

Step 3: Locate the block that contains the host address.

Step 4: Network ID is the first address; broadcast is the last.

Step 5: Usable hosts are between network and broadcast addresses.

Read the data before changing it

Command Prompt — Sales2

```
C:\> ipconfig /all

Ethernet adapter Ethernet:
   Description . . . . . . . . . : Ethernet Network Adapter
   IPv4 Address. . . . . . . . . : 192.168.1.12
   Subnet Mask . . . . . . . . . : 255.255.255.224
   Default Gateway . . . . . . . : 192.168.1.1

C:\> ping 192.168.1.1
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
```

Useful commands

ipconfig /all	Show full adapter IP addressing information.
route print	Inspect the default route and subnet routes.
ping <gateway>	Test the configured Layer 3 next hop.
ping <host>	Test reachable network endpoints.

Troubleshooting tip: If the switch link shows UP but an inter-subnet ping fails, inspect the subnet mask and each computer's gateway. Do not assume a cable creates IP routing.

Essential vocabulary

IP address	Layer 3 logical identifier for an interface
Subnet mask	Separates network bits from host bits
CIDR	Slash prefix expressing the network mask
Network ID	First address in a subnet; identifies the subnet
Broadcast	Last address used for IPv4 local broadcast
Usable host	Address between the network and broadcast
Default gateway	Router used to reach other IP networks
Router	Forwards packets between distinct IP subnets
Switch	Forwards Ethernet frames based on MAC addresses
Broadcast domain	Devices sharing Layer 2 broadcast delivery
IP configuration	Host IP address, mask, and gateway settings
ipconfig /all	Windows command for adapter addressing details
Ping / ICMP	Common tool used to test reachability
Block size	Number of IPv4 addresses in each subnet

Review before starting the HTML lab

1. Which /27 network contains 192.168.1.42?
2. What are the first and last usable IPs in 192.168.1.64/27?
3. Why can a Layer 2 switch not route from Sales to Accounting?
4. Which Windows command lists the subnet mask?